

The Deutsch–Jozsa Algorithm

The first algorithm with a provable quantum advantage. One oracle query decides what a classical machine needs $2^{n-1}+1$ queries to decide.

EXAMPLE NOTES

Written by the QuantaVerse team to show what a module's notes can look like: each lesson's theory in brief, every formula typeset and numbered, and the key ones gathered on a formula sheet at the end. Professors can upload their own notes for this module on its page, and students will see them alongside these.

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CONVENTIONS

x and z are n -bit strings, written $x_{n-1} \cdots x_1 x_0$ as Qiskit prints them, so 01 means $x_0 = 1$. $\oplus$ is XOR, and $x \cdot z = x_0 z_0 \oplus \cdots \oplus x_{n-1} z_{n-1}$ is the bitwise inner product mod 2. The input register has n qubits and the answer (ancilla) register one.

LESSON 6.1

The problem

KEY IDEA

A function is promised to be either constant or balanced, and the task is to tell which. A deterministic classical algorithm needs, in the worst case, just over half of all inputs. The quantum algorithm needs one query.

$$f : \{0,1\}^n \rightarrow \{0,1\}, \quad \text{constant: } f(x) = c \ \forall x, \quad \text{balanced: } |f^{-1}(0)| = |f^{-1}(1)| = 2^{n-1} \quad (6.1.1)$$

the promise

Classically, each query reveals one value $f(x)$. Seeing two different values proves “balanced” at once, but seeing only equal values proves nothing until more than half of the inputs have been checked: a balanced function can hide all its 1s in the half not yet asked about.

$$Q_{\text{classical, deterministic}} = 2^{n-1} + 1, \quad Q_{\text{quantum}} = 1 \quad (6.1.2)$$

worst-case queries

$$\#\{\text{constant } f\} = 2, \quad \#\{\text{balanced } f\} = \binom{2^n}{2^{n-1}} \quad (6.1.3)$$

how many functions satisfy the promise

The case $n = 1$ is Deutsch’s original problem: is $f(0) = f(1)$? Classically that takes two evaluations, and the quantum circuit takes one.

n	INPUTS 2^n	CLASSICAL WORST CASE	QUANTUM
1	2	2	1
3	8	5	1
10	1024	513	1
50	$\approx 1.1 \times 10^{15}$	$\approx 5.6 \times 10^{14}$	1

A randomised classical algorithm does far better than the worst case. If f is balanced, k random queries all return the same value with probability at most 2^{1-k} , so a few queries give high confidence. The honest summary is an *exact* separation, not a practical one.

$$P(\text{a balanced } f \text{ looks constant after } k \text{ random queries}) \leq 2^{1-k} \quad (6.1.4)$$

why randomness nearly closes the gap

WORKED EXAMPLE · $N = 2$, ALL SIX BALANCED FUNCTIONS

On inputs 00, 01, 10, 11 a balanced f has exactly two 1s: $\binom{4}{2} = 6$ choices, for example $f(x) = x_0$, $f(x) = x_1$ and $f(x) = x_0 \oplus x_1$. A classical algorithm that has seen $f(00) = f(01) = 0$ still cannot rule out $f = x_1$, so it must ask a third time.

LESSON 6.2

Oracles

KEY IDEA

A quantum gate must be reversible, but most functions are not. The standard fix keeps the input and XORs the output onto an extra qubit.

$$U_f|x\rangle|y\rangle = |x\rangle|y \oplus f(x)\rangle \quad (6.2.1)$$

the bit oracle

$$U_f^2 = I \quad \text{since} \quad y \oplus f(x) \oplus f(x) = y \quad (6.2.2)$$

every such oracle is its own inverse, whatever f is

The map $|x\rangle \mapsto |f(x)\rangle$ would not be a gate: two inputs with the same output cannot be told apart afterwards, so no unitary does it. Keeping x makes the map a permutation of basis states, which is always unitary.

FUNCTION	ORACLE CIRCUIT
$f(x) = 0$	nothing: the identity
$f(x) = 1$	X on the answer qubit
$f(x) = x_i$	CNOT from input i to the answer
$f(x) = x_0 \oplus x_1$	CNOT from input 0, then CNOT from input 1
$f(x) = s \cdot x$	one CNOT from each input i with $s_i = 1$

$$f(x) = s \cdot x \Rightarrow U_f = \prod_{i: s_i = 1} \text{CNOT}_{x_i \rightarrow y} \quad (6.2.3)$$

linear oracles are just CNOTs

When the answer qubit is prepared in $|-\rangle$, the bit oracle acts as a *phase oracle* on the input register alone (lesson 6.3). Many algorithms, Grover's included, are written with the phase form directly.

$$O_f|x\rangle = (-1)^{f(x)}|x\rangle \quad (6.2.4)$$

the phase oracle

DEFINITION · QUERY

One application of U_f or O_f . Query complexity counts only these calls and ignores the rest of the circuit. That is what makes separations provable, and also what limits what they say (lesson 6.5).

WORKED EXAMPLE · THE WALKTHROUGH'S ORACLE

For $f(x_0, x_1) = x_0 \oplus x_1$, applying CNOT from x_0 and then from x_1 gives $|x_0 x_1\rangle|y\rangle \mapsto |x_0 x_1\rangle|y \oplus x_0 \oplus x_1\rangle$. On $|10\rangle|0\rangle$ the answer qubit becomes 1, and on $|11\rangle|0\rangle$ it returns to 0: balanced, with two 1s among four inputs.

LESSON 6.3

Phase kickback

KEY IDEA

With the answer qubit in $|-\rangle$, the oracle leaves that qubit unchanged and multiplies the input term by $(-1)^{f(x)}$. The answer arrives as a sign.

$$U_f |x\rangle |-\rangle = (-1)^{f(x)} |x\rangle |-\rangle \quad (6.3.1)$$

phase kickback

$$U_f |x\rangle \frac{|0\rangle - |1\rangle}{\sqrt{2}} = |x\rangle \frac{|f(x)\rangle - |1 \oplus f(x)\rangle}{\sqrt{2}} = (-1)^{f(x)} |x\rangle \frac{|0\rangle - |1\rangle}{\sqrt{2}} \quad (6.3.2)$$

the derivation: if $f(x) = 1$ the two terms swap, which is a sign

On a single basis input the sign is a global phase and means nothing. On a superposition of inputs, every input picks up its own sign, which makes it a *relative* phase: the whole truth table of f written into the signs of one state.

$$U_f \left(\frac{1}{\sqrt{2^n}} \sum_x |x\rangle \right) |-\rangle = \left(\frac{1}{\sqrt{2^n}} \sum_x (-1)^{f(x)} |x\rangle \right) |-\rangle \quad (6.3.3)$$

one query, every value of f , stored as signs

The answer qubit is prepared from $|0\rangle$ by X then H . It ends in $|-\rangle$ again, unentangled from the input register, so it can be ignored.

$$HX|0\rangle = H|1\rangle = |-\rangle \quad (6.3.4)$$

preparing the ancilla

DEUTSCH-JOZSA, $N = 2$



WORKED EXAMPLE · SIGNS FOR $F = x_0 \oplus x_1$

After the first Hadamards the inputs are $\frac{1}{2}(|00\rangle + |01\rangle + |10\rangle + |11\rangle)$.

The oracle multiplies each term by $(-1)^{x_0 \oplus x_1}$, giving $\frac{1}{2}(|00\rangle - |01\rangle - |10\rangle + |11\rangle)$. The probabilities are still a quarter each, so the information is present but not yet readable.

LESSON 6.4

Interference

KEY IDEA

A second layer of Hadamards turns the pattern of signs into amplitudes. Every path to $|0 \cdots 0\rangle$ adds up when f is constant and cancels exactly when f is balanced.

$$H^{\otimes n}|x\rangle = \frac{1}{\sqrt{2^n}} \sum_z (-1)^{x \cdot z} |z\rangle \quad (6.4.1)$$

the Hadamard transform of a basis state

$$|\psi_{\text{final}}\rangle = \sum_z \left[\frac{1}{2^n} \sum_x (-1)^{f(x) + x \cdot z} \right] |z\rangle \quad (6.4.2)$$

the input register just before measurement

The amplitude of the all-zero string has $x \cdot 0 = 0$ in every term, so it is simply the average of the signs.

$$\alpha_{0 \cdots 0} = \frac{1}{2^n} \sum_x (-1)^{f(x)} = \begin{cases} \pm 1 & f \text{ constant} \\ 0 & f \text{ balanced} \end{cases} \quad (6.4.3)$$

the decision rule

A constant f puts all the amplitude on $|0 \cdots 0\rangle$, so measurement returns all zeros with certainty. A balanced f has exactly as many $+1$ as -1 terms, so all zeros never appears, and any other result means balanced. No probability is involved in either case.

MEASURED STRING	CONCLUSION	CERTAINTY
$0 \cdots 0$	constant	exact
anything else	balanced	exact

WORKED EXAMPLE · FINISHING F = $x_0 \oplus x_1$

For $z = 11$: $x \cdot z = x_0 \oplus x_1 = f(x)$, so every exponent $f(x) + x \cdot z$ is even and

$$\alpha_{11} = \frac{1}{4}(1 + 1 + 1 + 1) = 1.$$

The measurement returns 11 every time. That is not all zeros, so f is balanced, decided with one query. The walkthrough on the algorithms page shows the same four signs combining.

In general the balanced case can spread its amplitude over many strings. Only the zero string is guaranteed to be empty, and that is all the algorithm needs.

LESSON 6.5

Query complexity

KEY IDEA

Deutsch–Jozsa proves an exponential separation between exact quantum and deterministic classical query complexity. It does not prove a practical speed-up, because randomised classical algorithms need only a few queries.

MODEL	QUERIES	ERROR
deterministic classical	$2^{n-1} + 1$	none
randomised classical	$O(\log 1/\varepsilon)$	ε
quantum (Deutsch–Jozsa)	1	none

The same circuit solves a sibling problem with a more useful separation. In *Bernstein–Vazirani* the oracle computes $f(x) = s \cdot x$ for a hidden string s , and one query returns all of s . Classically each query yields at most one bit of s , so n queries are needed even with randomness.

$$f(x) = s \cdot x \Rightarrow H^{\otimes n} O_f H^{\otimes n} |0\rangle^{\otimes n} = |s\rangle \quad (6.5.1)$$

Bernstein–Vazirani: the hidden string, read in one query

$$\frac{1}{2^n} \sum_x (-1)^{s \cdot x + z \cdot x} = \frac{1}{2^n} \sum_x (-1)^{(s \oplus z) \cdot x} = \begin{cases} 1 & z = s \\ 0 & z \neq s \end{cases} \quad (6.5.2)$$

why every other string cancels

Simon’s problem goes further: a function hides a string s with $f(x) = f(y) \Leftrightarrow y \in \{x, x \oplus s\}$. A quantum algorithm needs $O(n)$ queries, while any classical algorithm, randomised or not, needs about $2^{n/2}$. Simon’s algorithm directly inspired Shor’s (module 8).

$$Q_{\text{Simon, quantum}} = O(n), \quad Q_{\text{Simon, classical}} = \Omega(2^{n/2}) \quad (6.5.3)$$

an exponential separation that randomness cannot close

DEFINITION · ORACLE SEPARATION

A proof about how many calls to a black box are needed. It says nothing about any function written out explicitly, where a classical algorithm may read the circuit and skip the queries entirely.

WORKED EXAMPLE · BERNSTEIN–VAZIRANI, $n = 2$

With $s = 01$ (so $f(x) = x_0$), the walkthrough’s oracle is one CNOT from x_0 . After the final Hadamards the register reads $x_1 x_0 = 01$ with certainty: the secret, from a single query, where two classical queries (inputs 01 and 10) would be needed.

MODULE 6 · FORMULA SHEET

Formula sheet

The key results of the module, numbered as they appear in the lessons.

the promise	$f : \{0,1\}^n \rightarrow \{0,1\}$, constant: $f(x) = c \ \forall x$, balanced: $ f^{-1}(0) = f^{-1}(1) = 2^{n-1}$	(6.1.1)
worst-case queries	$Q_{\text{classical, deterministic}} = 2^{n-1} + 1$, $Q_{\text{quantum}} = 1$	(6.1.2)
the bit oracle	$U_f x\rangle y\rangle = x\rangle y \oplus f(x)\rangle$	(6.2.1)
every such oracle is its own inverse, whatever f is	$U_f^2 = I$ since $y \oplus f(x) \oplus f(x) = y$	(6.2.2)
the phase oracle	$O_f x\rangle = (-1)^{f(x)} x\rangle$	(6.2.4)
phase kickback	$U_f x\rangle -\rangle = (-1)^{f(x)} x\rangle -\rangle$	(6.3.1)
one query, every value of f , stored as signs	$U_f\left(\frac{1}{\sqrt{2^n}} \sum_x x\rangle\right) -\rangle = \left(\frac{1}{\sqrt{2^n}} \sum_x (-1)^{f(x)} x\rangle\right) -\rangle$	(6.3.3)
the Hadamard transform of a basis state	$H^{\otimes n} x\rangle = \frac{1}{\sqrt{2^n}} \sum_z (-1)^{x \cdot z} z\rangle$	(6.4.1)
the input register just before measurement	$ \psi_{\text{final}}\rangle = \sum_z \left[\frac{1}{2^n} \sum_x (-1)^{f(x) + x \cdot z} \right] z\rangle$	(6.4.2)
the decision rule	$\alpha_0 \dots \alpha_n = \frac{1}{2^n} \sum_x (-1)^{f(x)} = \begin{cases} \pm 1 & f \text{ constant} \\ 0 & f \text{ balanced} \end{cases}$	(6.4.3)
Bernstein–Vazirani: the hidden string, read in one query	$f(x) = s \cdot x \Rightarrow H^{\otimes n} O_f H^{\otimes n} 0\rangle^{\otimes n} = s\rangle$	(6.5.1)
an exponential separation that randomness cannot close	$Q_{\text{Simon, quantum}} = O(n)$, $Q_{\text{Simon, classical}} = \Omega(2^{n/2})$	(6.5.3)